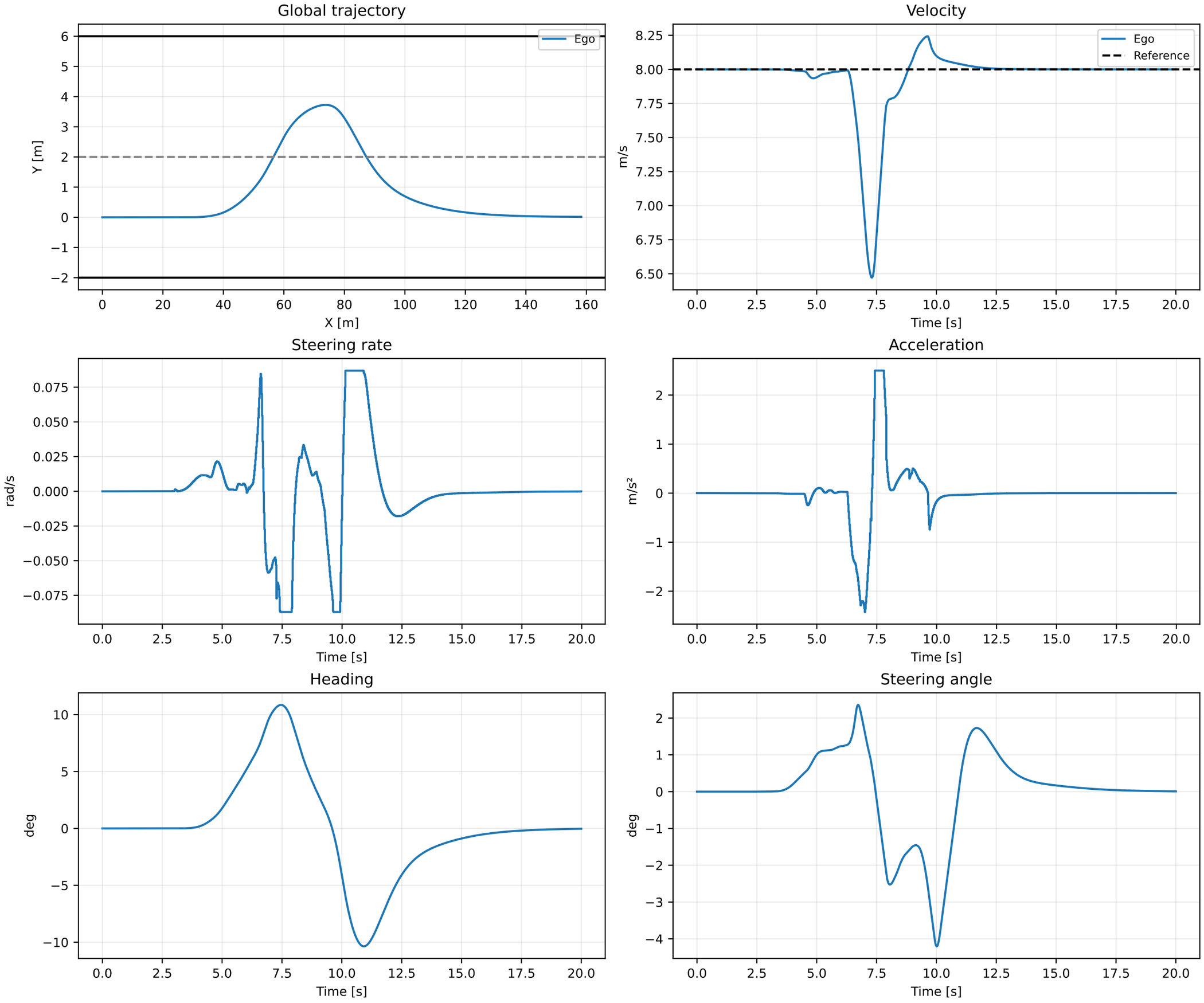
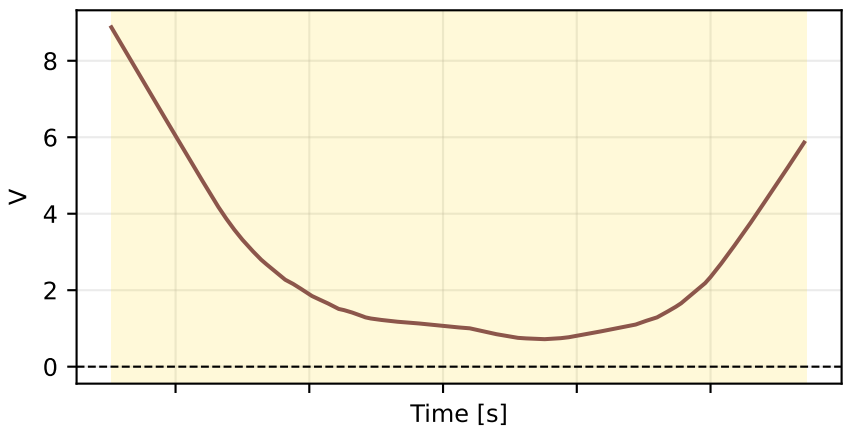
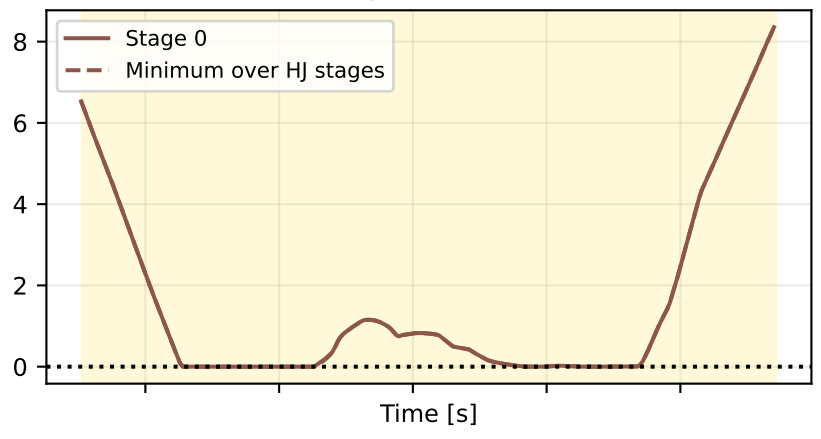

Global trajectory
Ego
Y [m]
X [m]
Velocity
Ego
Reference
m/s
Time [s]
Steering rate
rad/s
Time [s]
Acceleration
m/s²
Time [s]
Heading
deg
Time [s]
Steering angle
deg
Time [s]

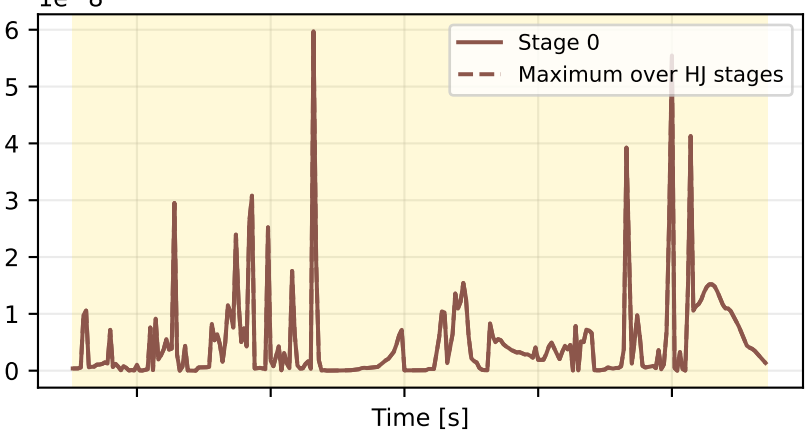
btn: BRT



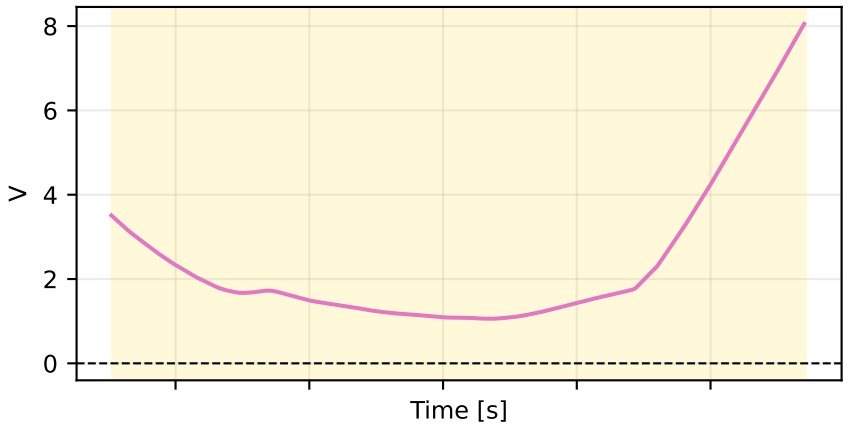
HJ residual



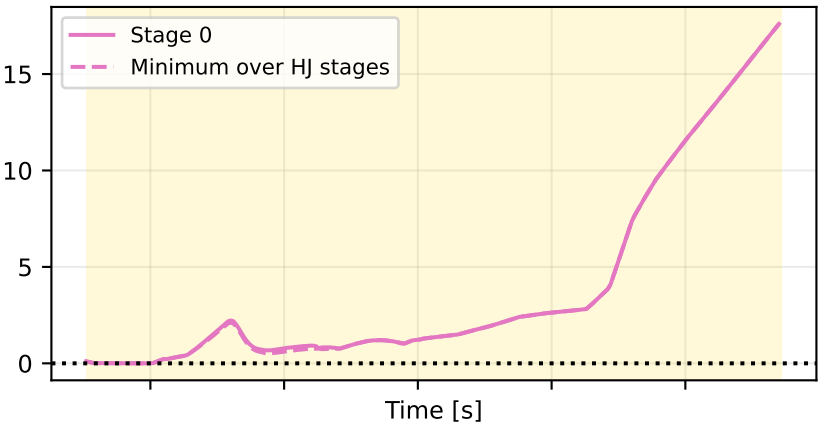
Lower slack



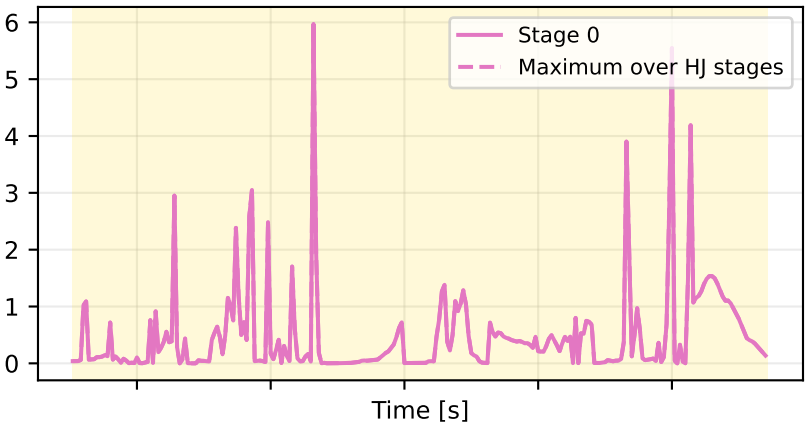
stn: BRT



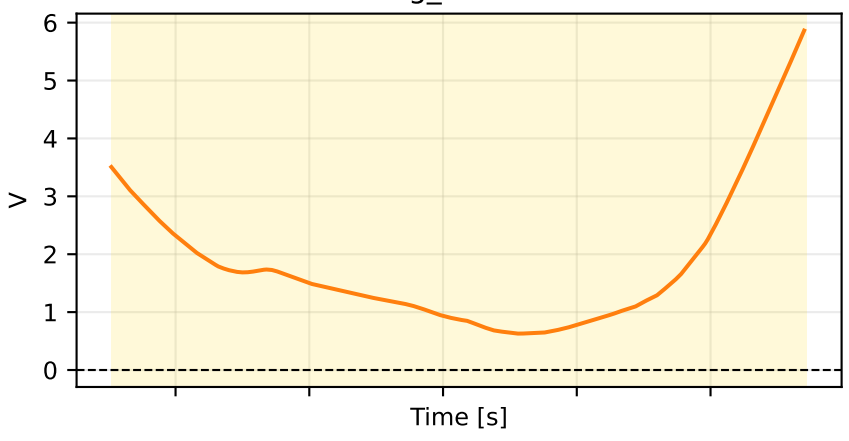
HJ residual



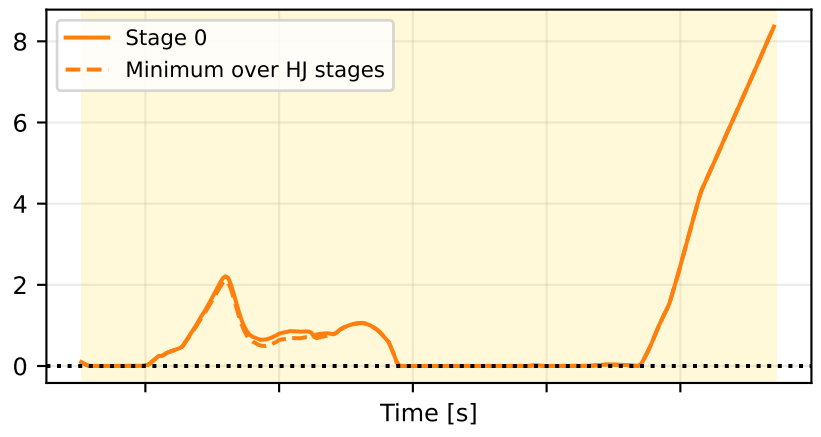
Lower slack



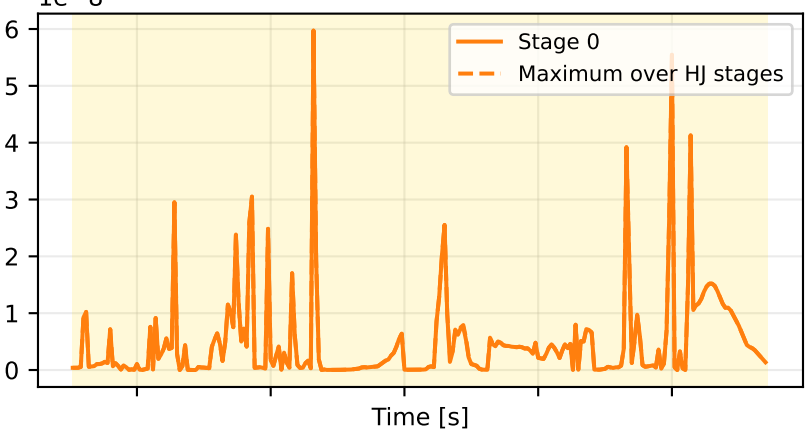
long_lat: BRT



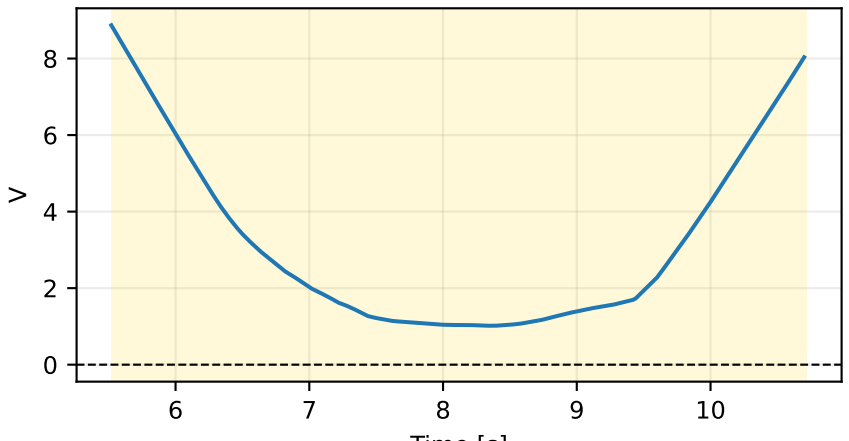
HJ residual



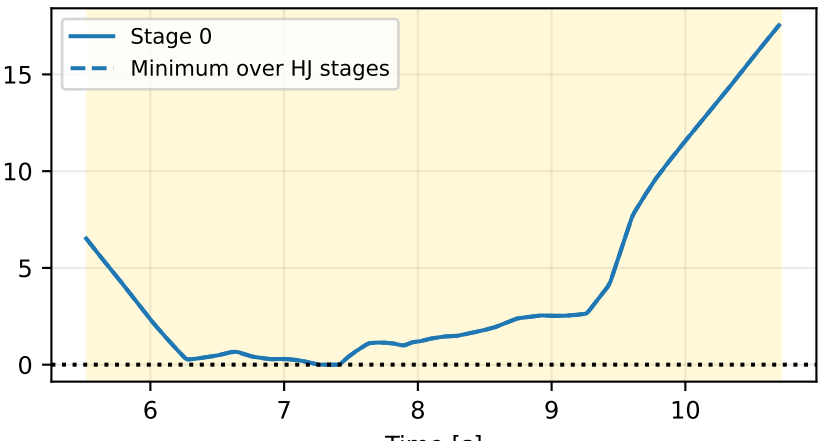
Lower slack



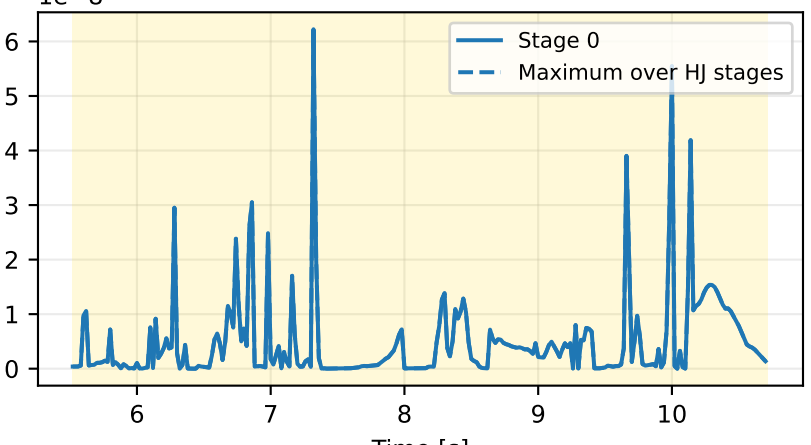
euclidean: BRT

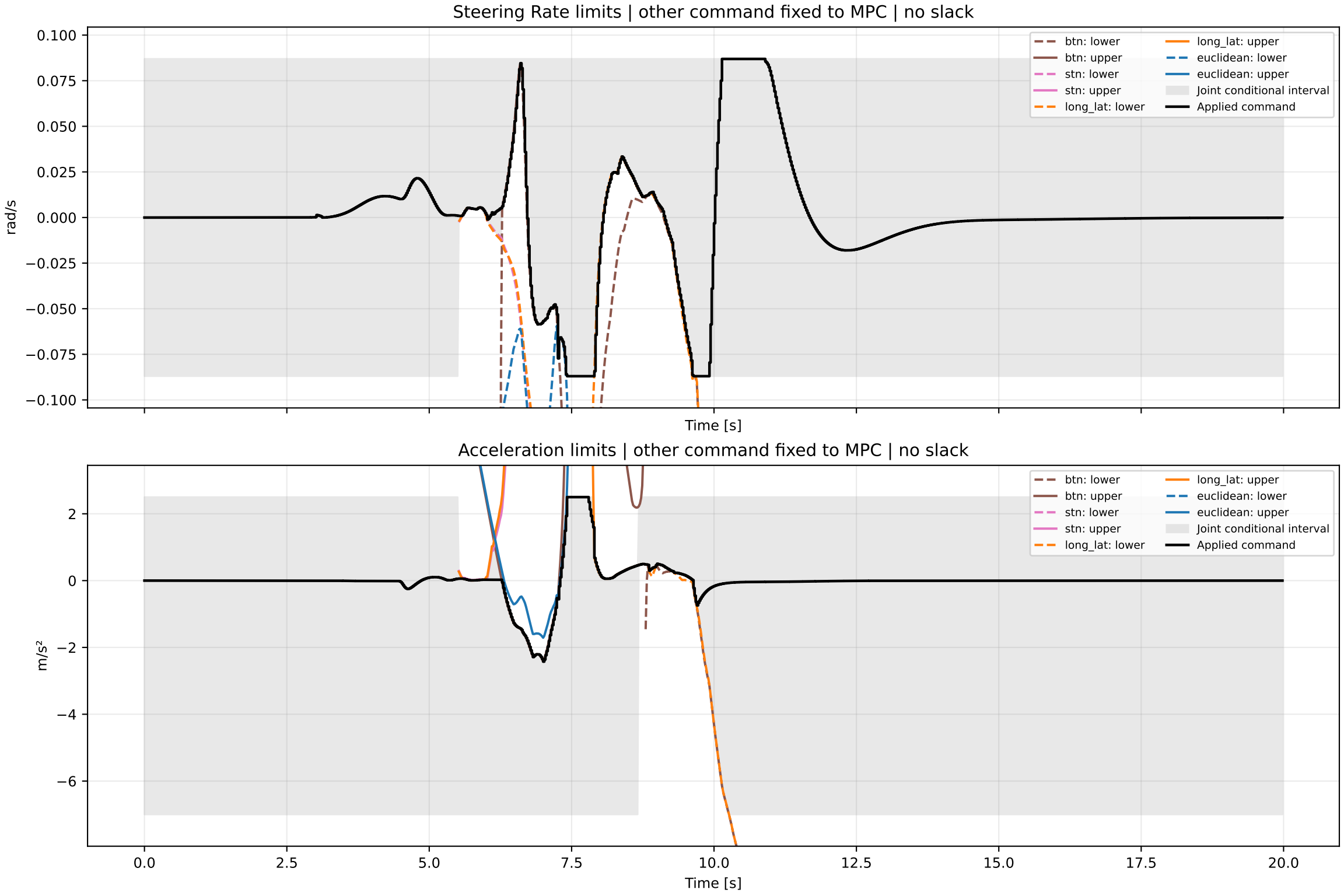


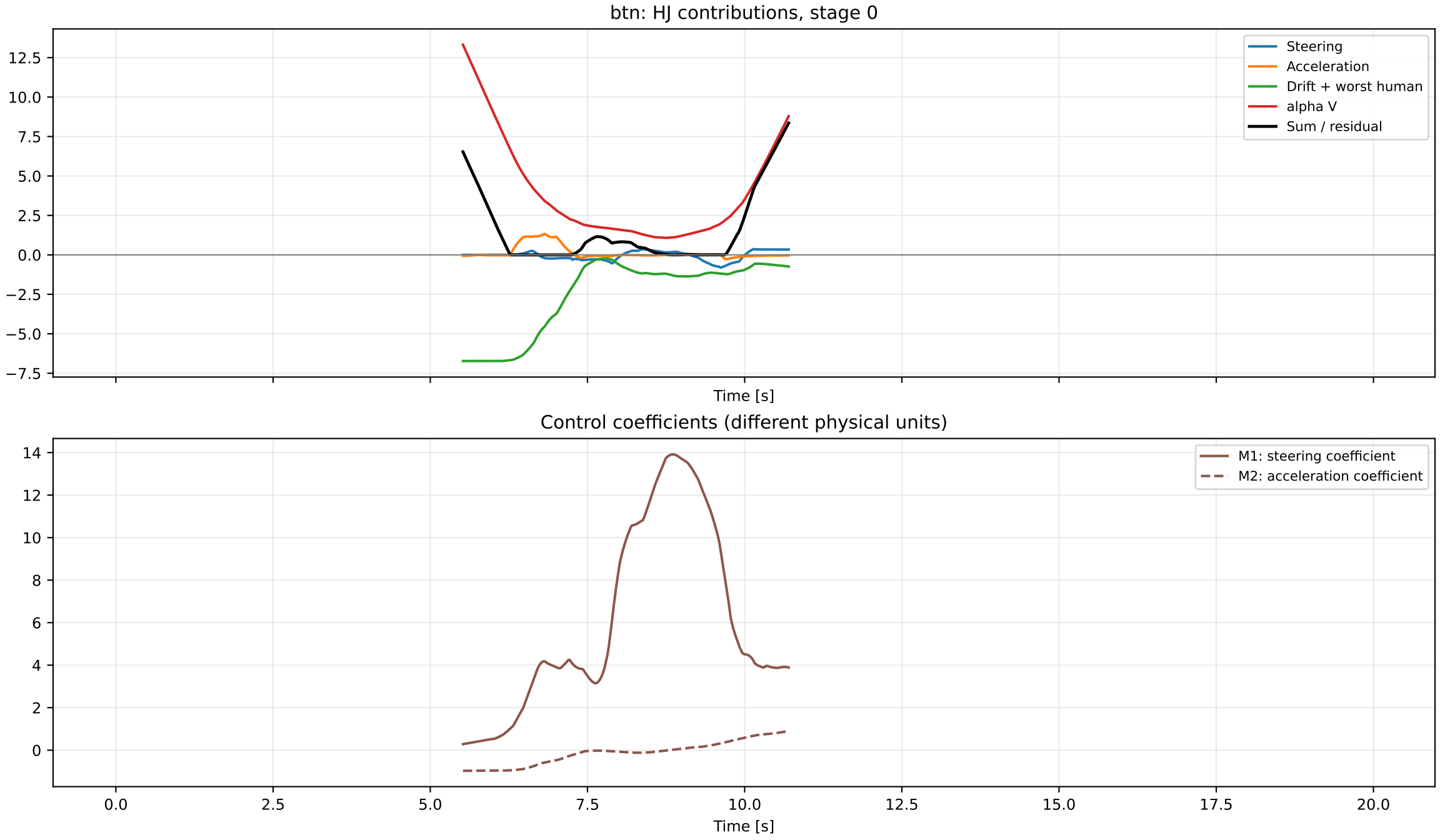
HJ residual

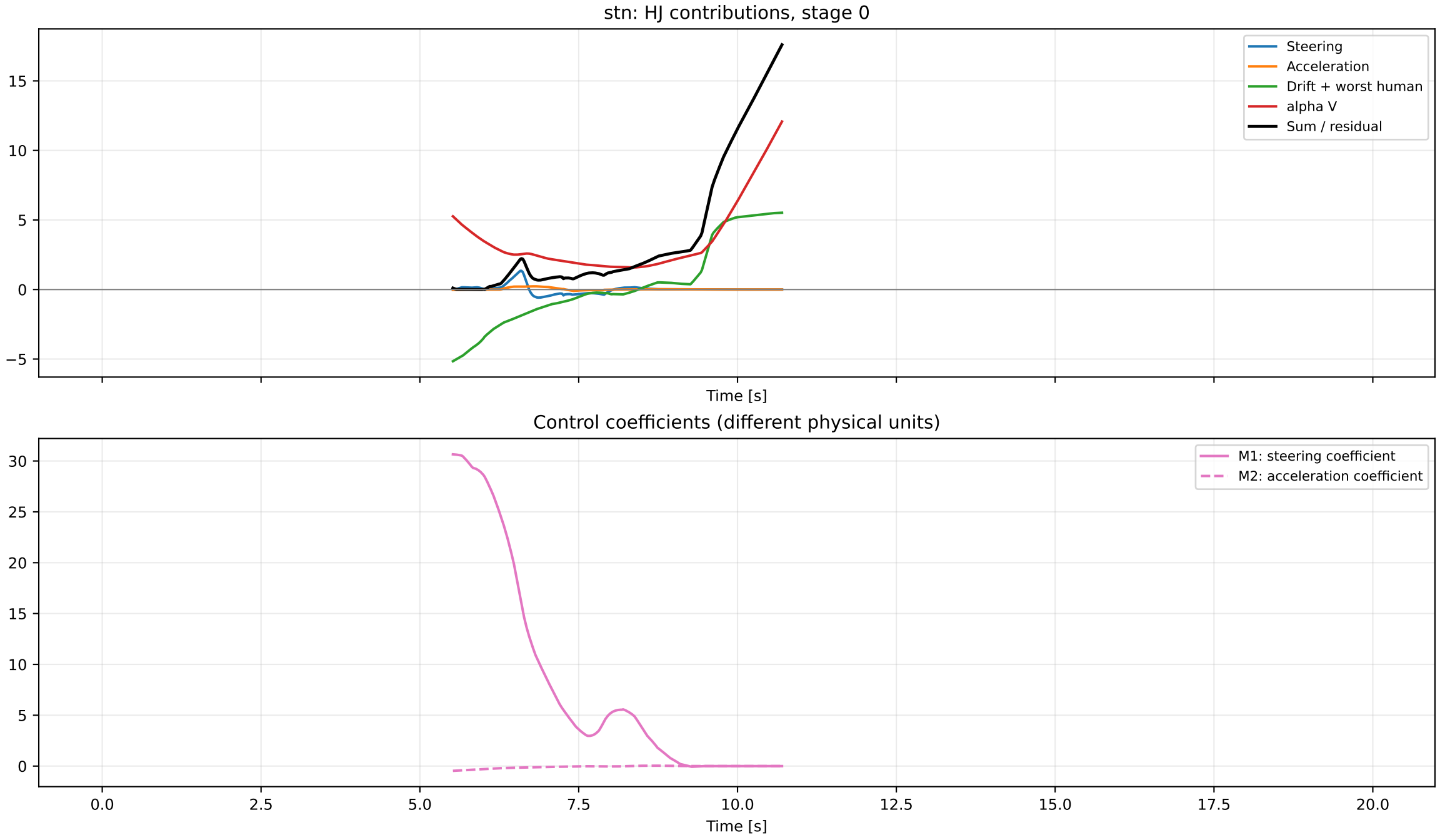


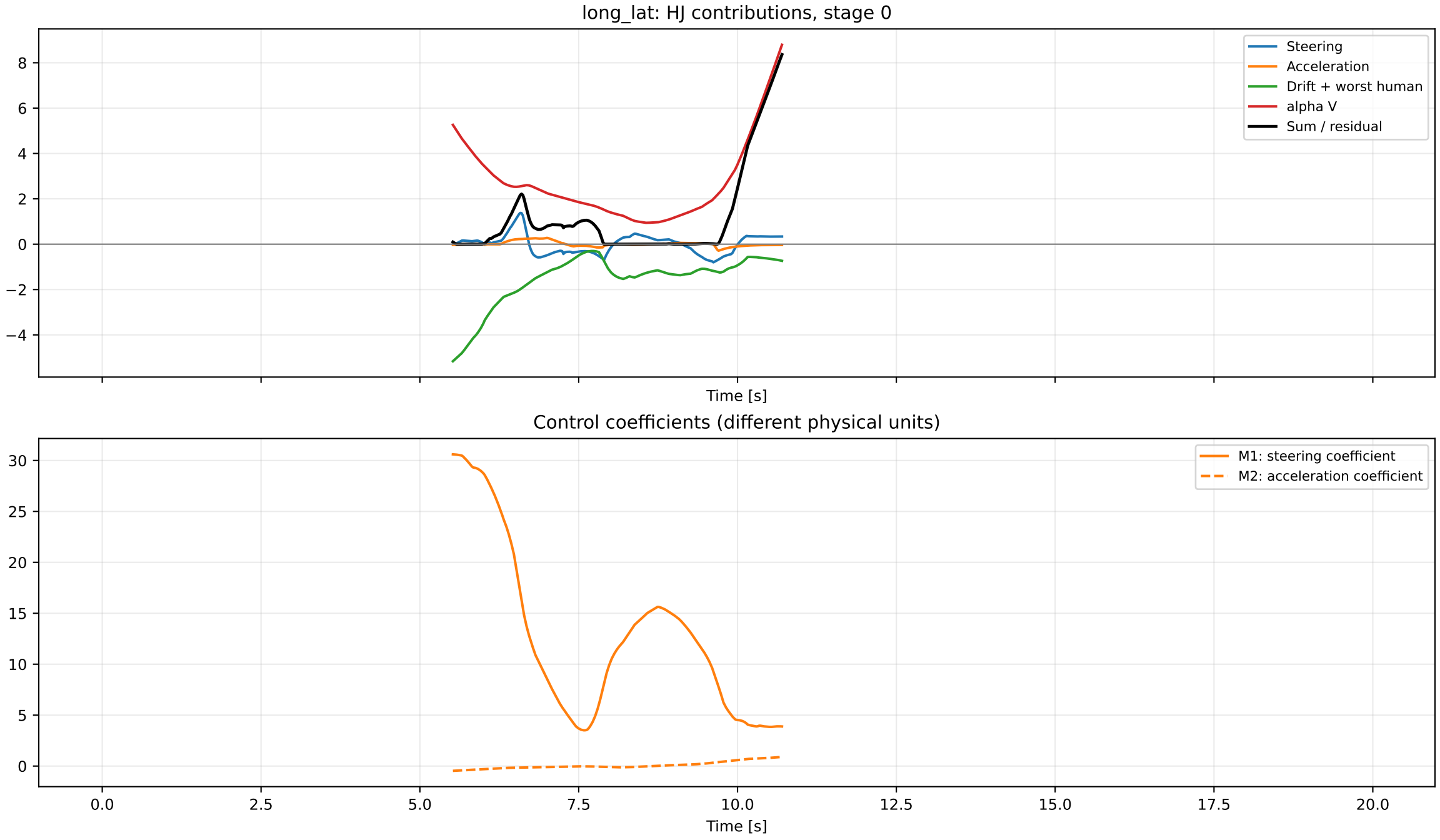
Lower slack

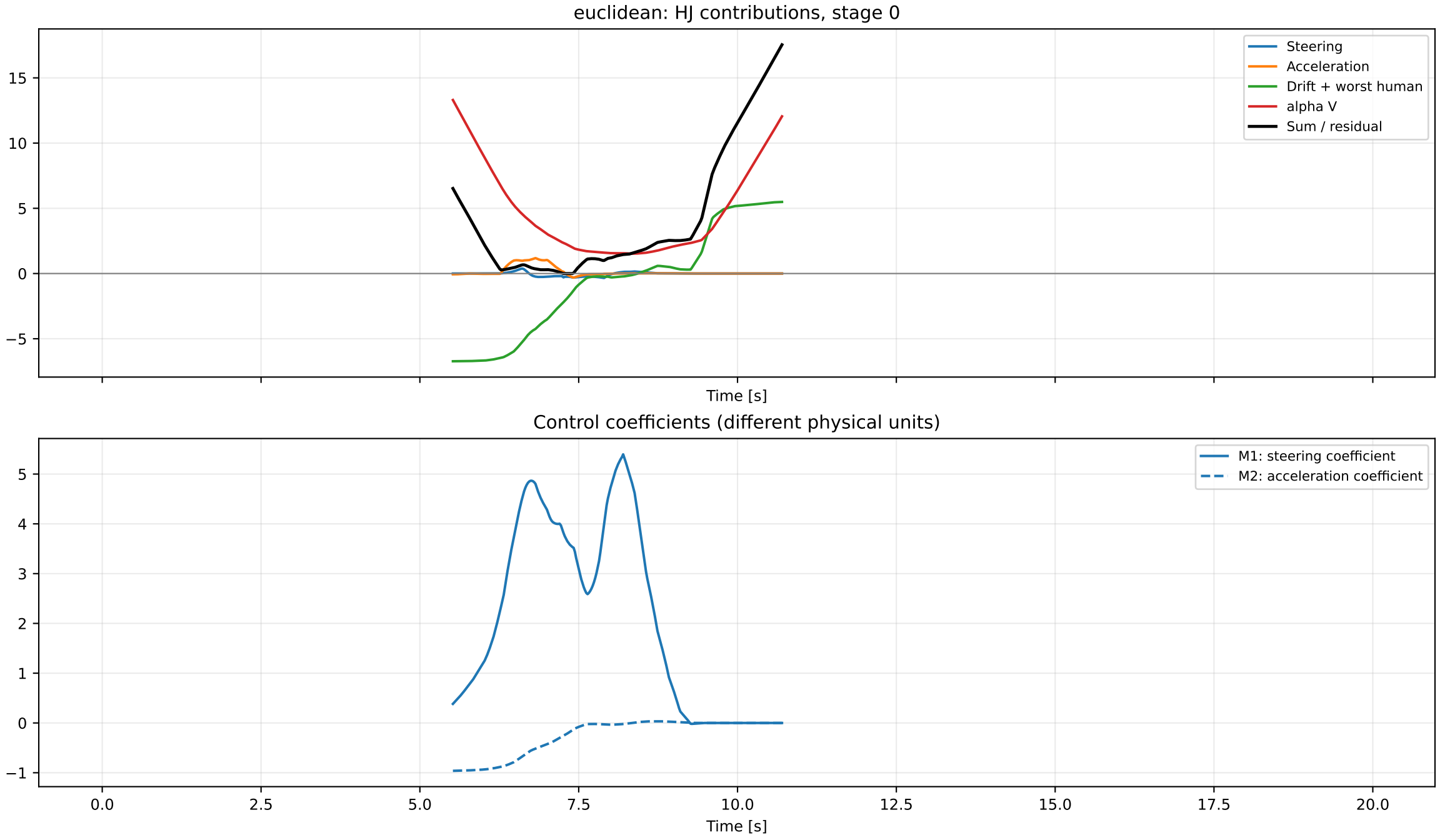


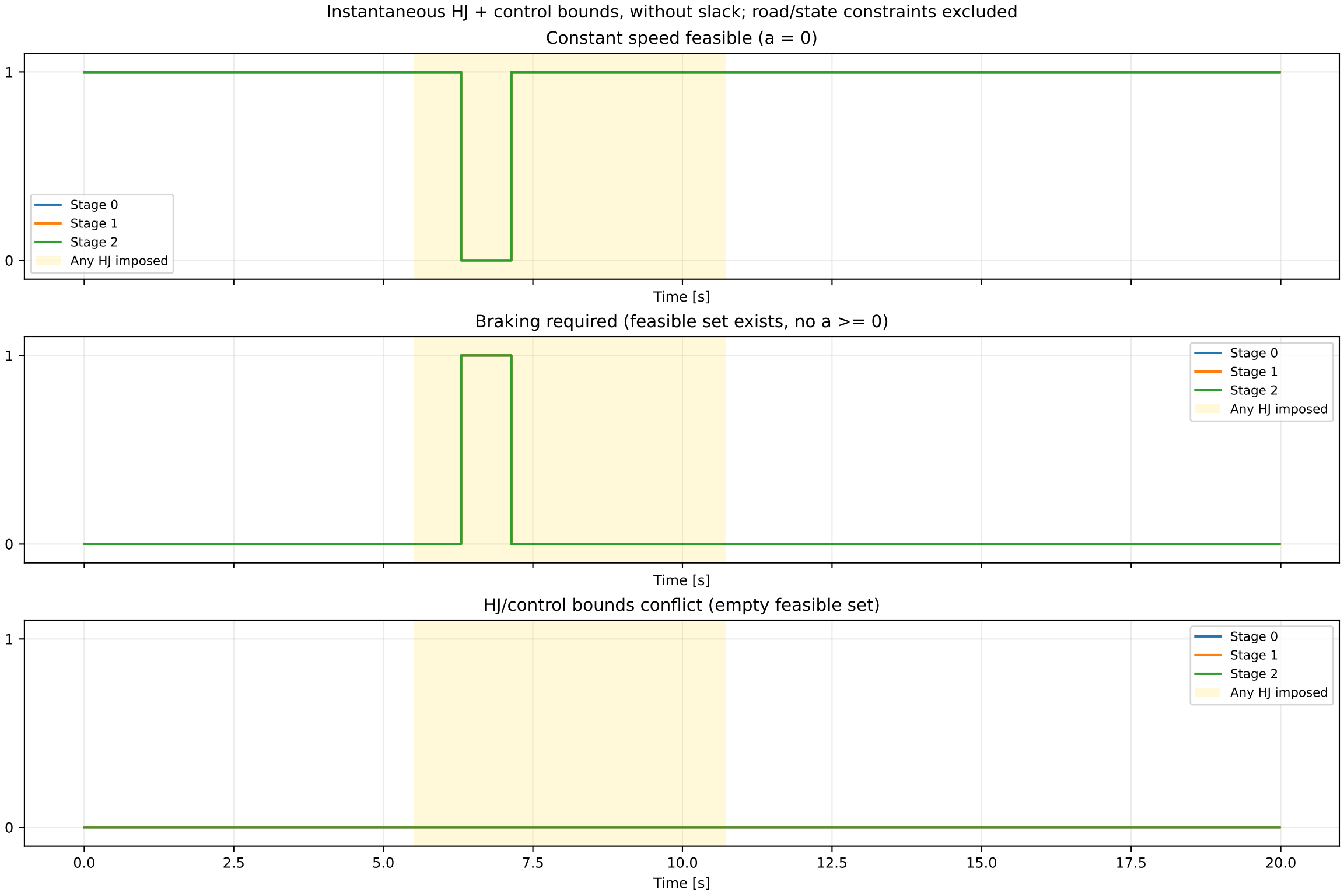

Steering Rate limits | other command fixed to MPC | no slack
rad/s
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
Time [s]
btn: lower
btn: upper
stn: lower
stn: upper
long_lat: lower
long_lat: upper
euclidean: lower
euclidean: upper
Joint conditional interval
Applied command
Acceleration limits | other command fixed to MPC | no slack
m/s²
2
0
−2
−4
−6
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

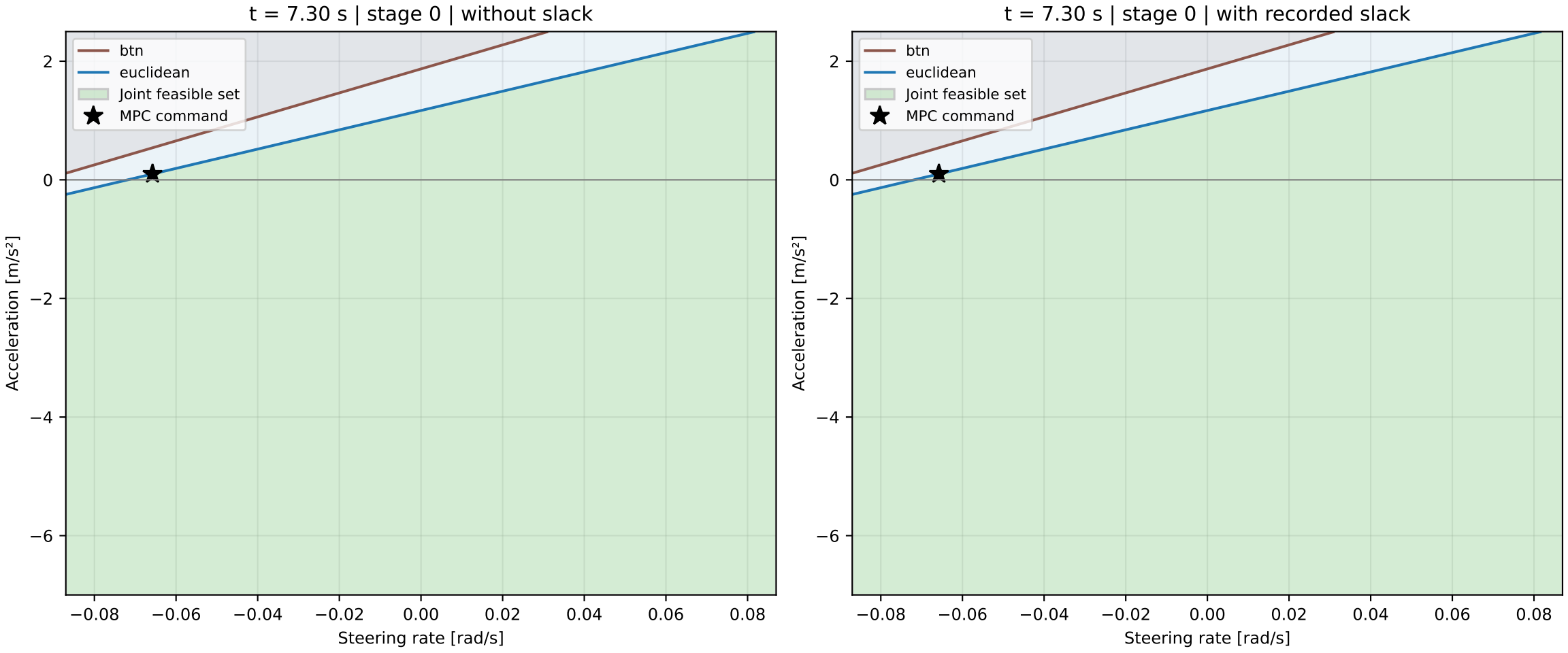

btn: HJ contributions, stage 0
Steering
Acceleration
Drift + worst human
alpha V
Sum / residual
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
Time [s]
Control coefficients (different physical units)
M1: steering coefficient
M2: acceleration coefficient
14
12
10
8
6
4
2
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

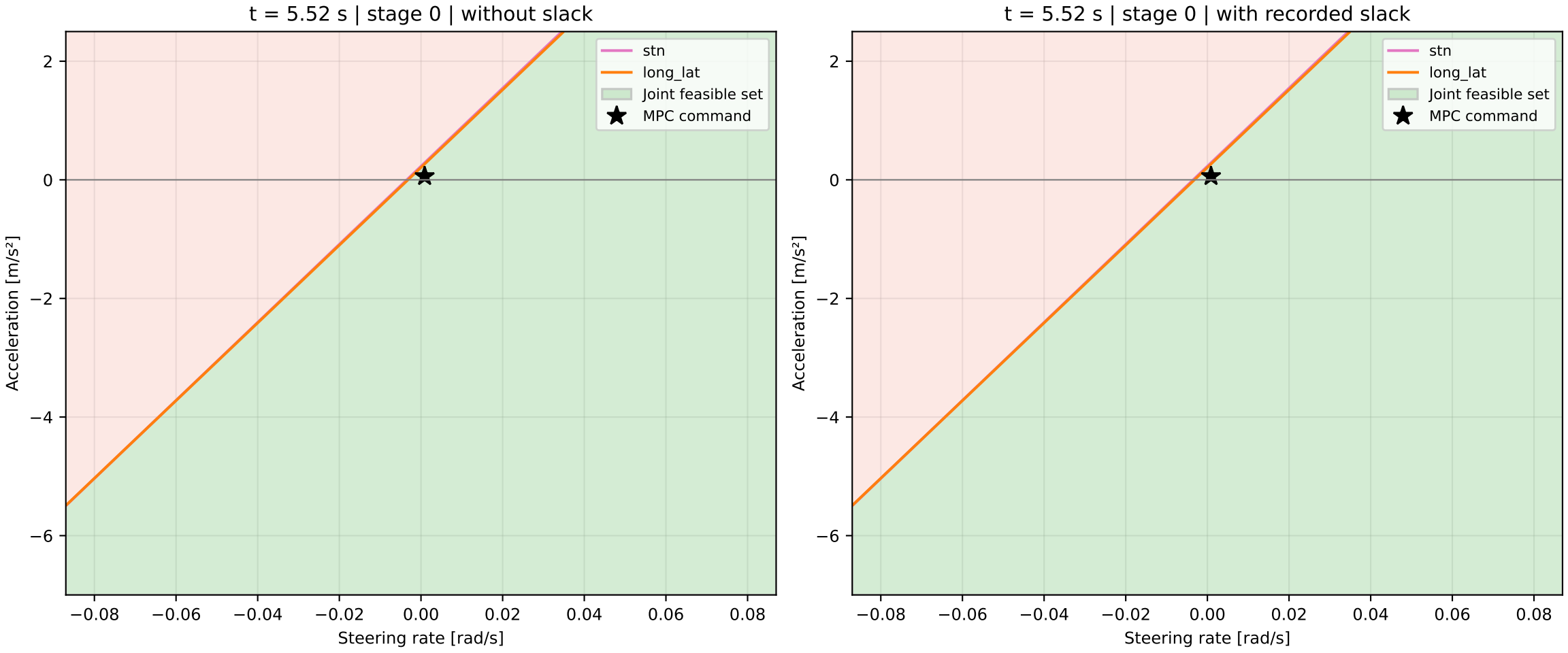

stn: HJ contributions, stage 0
Steering
Acceleration
Drift + worst human
alpha V
Sum / residual
15
10
5
0
−5
Time [s]
Control coefficients (different physical units)
M1: steering coefficient
M2: acceleration coefficient
30
25
20
15
10
5
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

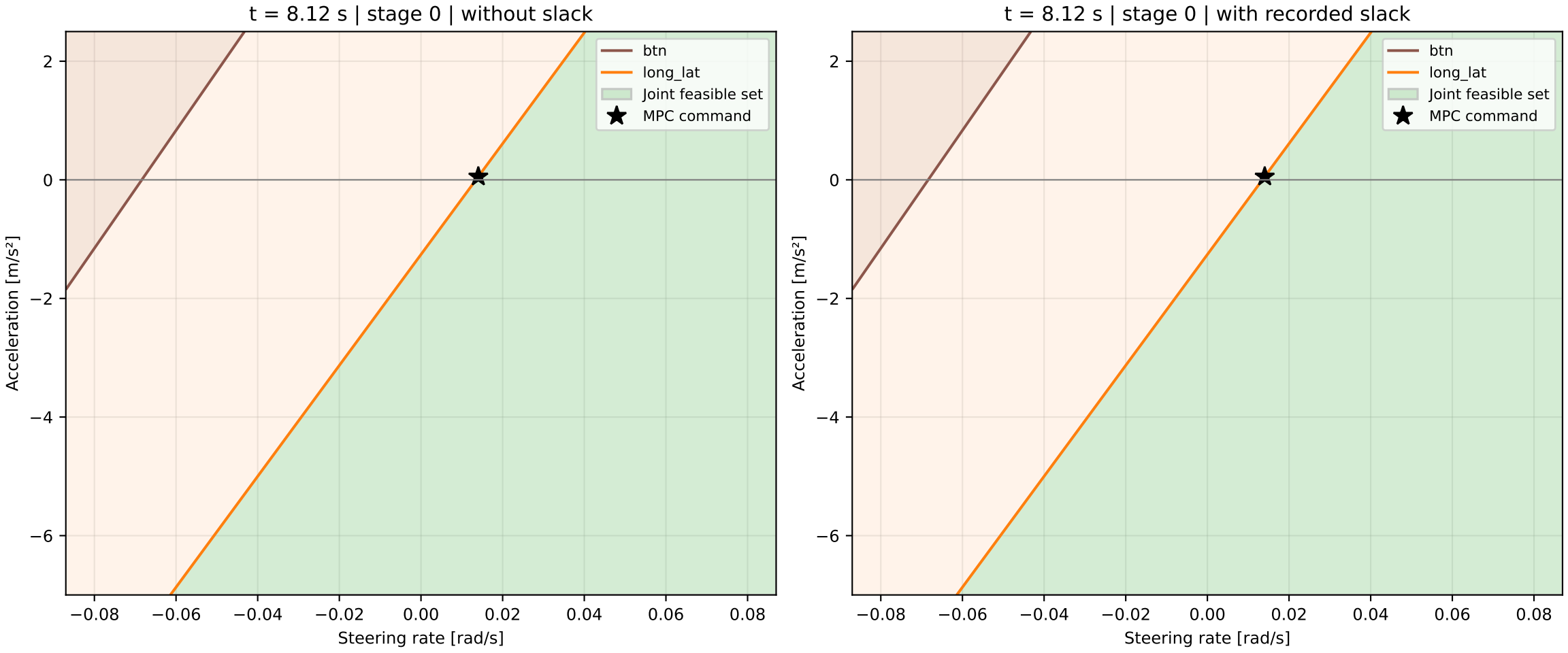

long_lat: HJ contributions, stage 0
Steering
Acceleration
Drift + worst human
alpha V
Sum / residual
8
6
4
2
0
−2
−4
Time [s]
Control coefficients (different physical units)
M1: steering coefficient
M2: acceleration coefficient
30
25
20
15
10
5
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

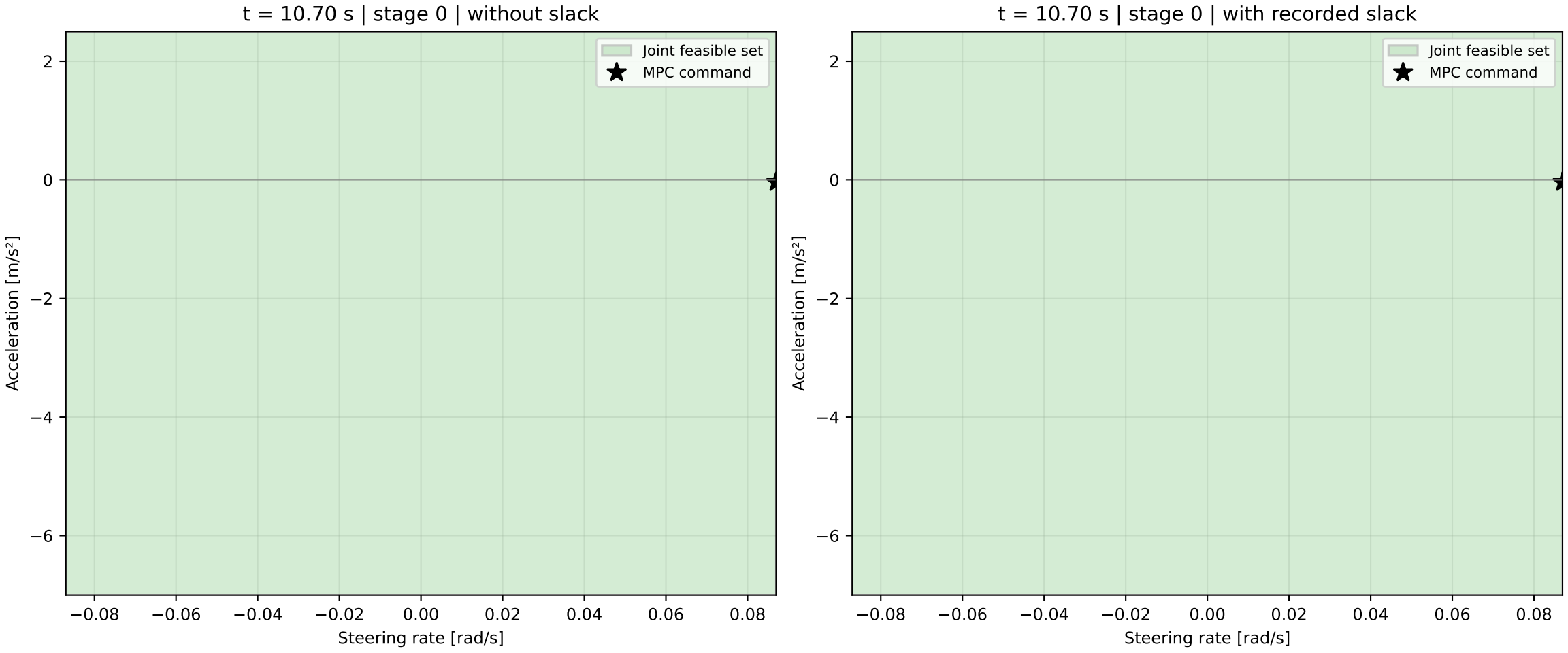

euclidean: HJ contributions, stage 0
Steering
Acceleration
Drift + worst human
alpha V
Sum / residual
15
10
5
0
−5
Time [s]
Control coefficients (different physical units)
M1: steering coefficient
M2: acceleration coefficient
5
4
3
2
1
0
−1
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

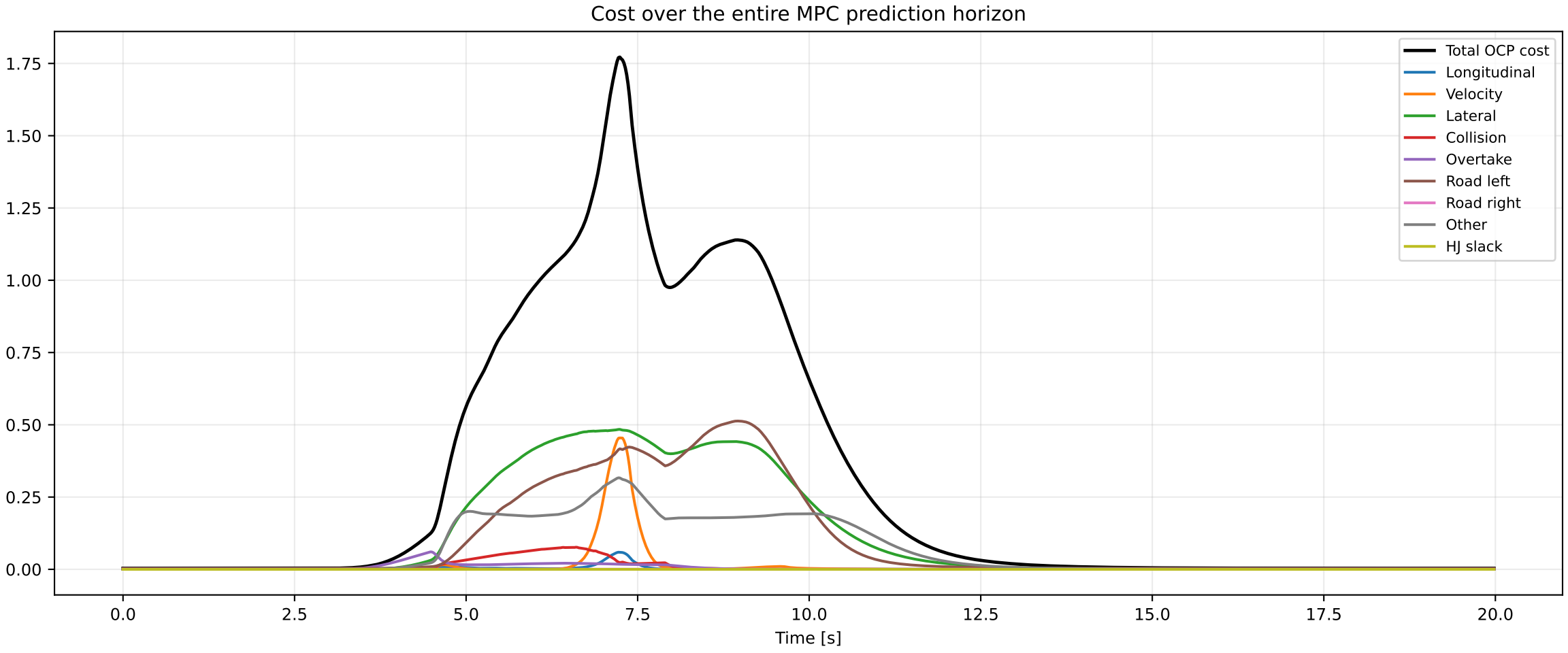

Instantaneous HJ + control bounds, without slack; road/state constraints excluded
Constant speed feasible (a = 0)
1
0
Stage 0
Stage 1
Stage 2
Any HJ imposed
Time [s]
Braking required (feasible set exists, no a >= 0)
1
0
Stage 0
Stage 1
Stage 2
Any HJ imposed
Time [s]
HJ/control bounds conflict (empty feasible set)
1
0
Stage 0
Stage 1
Stage 2
Any HJ imposed
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]


t = 7.30 s | stage 0 | without slack
t = 7.30 s | stage 0 | with recorded slack
btn
euclidean
Joint feasible set
MPC command
Acceleration [m/s²]
Steering rate [rad/s]
2
0
−2
−4
−6
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08


t = 5.52 s | stage 0 | without slack
t = 5.52 s | stage 0 | with recorded slack
stn
long_lat
Joint feasible set
MPC command
Acceleration [m/s²]
Steering rate [rad/s]
2
0
−2
−4
−6
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08


t = 8.12 s | stage 0 | without slack
t = 8.12 s | stage 0 | with recorded slack
btn
long_lat
Joint feasible set
MPC command
Acceleration [m/s²]
Steering rate [rad/s]
2
0
−2
−4
−6
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08


t = 10.70 s | stage 0 | without slack
t = 10.70 s | stage 0 | with recorded slack
Joint feasible set
MPC command
Acceleration [m/s²]
Steering rate [rad/s]
2
0
−2
−4
−6
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08


Cost over the entire MPC prediction horizon
Total OCP cost
Longitudinal
Velocity
Lateral
Collision
Overtake
Road left
Road right
Other
HJ slack
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]